



Project Report

Skill Swap & Student Service Exchange Platform

Course Title: Database Management System Lab

Course Code: CSE 224

Submitted to

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1. Executive Summary

SkillSwap is a web-based, peer-to-peer educational platform designed to allow students to exchange skills directly without financial barriers. The platform integrates a dynamic skill-listing system, request handling for skill trading, and an automated skill verification engine that tests students and assigns proficiency tiers (Beginner, Intermediate, Expert). An administrative portal provides oversight by tracking test performances, site analytics, and user deletion requests.

2. Group Details & Team Contributions

2.1 Group Details

- **Course:** Database Management Systems
- **Group Number:** Group 04

2.2 Team Member Contributions

Name	Role / Focus Area	Specific Contributions
MD. Nahidul Islam	Backend & PHP Admin	Developed backend logic, session management, dynamic quiz processing engine, and admin dashboard control panel (take_test.php, admin/dashboard.php).
Nahida Islam Shormi	Database Architecture & SQL	Designed relational database schema, written data queries, key constraints (FOREIGN KEY with CASCADE), and SQL seed scripts (schema.sql).
Abdullah As Sayef Tanzim	User Interface Styling (CSS)	Designed application UI layouts, color palettes, responsive card structures, forms, and custom stylesheet components.
Anika Tasnim	Frontend Markup (HTML)	Structured semantic page elements, navigational structures, forms, tables, and front-facing interface layout across all pages.

3. Introduction & Problem Statement

3.1 Introduction

In academic environments, students often possess strong practical skills in areas like programming, languages, or design, while lacking expertise in others. Traditional tutoring or online courses can be expensive or inaccessible. SkillSwap addresses this by enabling collaborative peer learning where knowledge serves as the currency.

3.2 Problem Statement

- **Financial Barriers:** Professional skill learning platforms require subscriptions or tuition fees.
- **Lack of Trust & Verification:** On informal platforms, students cannot verify whether a potential peer tutor actually possesses the skill level they claim.

- **Unstructured Peer Learning:** Without a structured request/approval lifecycle, finding compatible skill-swap partners remains inefficient.

4. System Objectives & Scope

4.1 Primary Objective

To build a scalable, secure, and user-friendly web platform that facilitates direct peer skill exchanges and provides automated skill verification through testing.

4.2 System Scope

- User authentication and role-based access control (Student vs. Admin).
- Skill management (CRUD operations for skills offered and skills wanted).
- Interactive 30-question assessment quizzes supporting single-choice and multiple-choice formats.
- Automated grading and dynamic tier assignment based on quiz scores.
- Admin dashboard for monitoring users, viewing verification test logs, and handling account deletion requests.

5. Implemented Features

5.1 User Authentication & Profile System

- **Secure Login & Registration:** Password-based account creation with session handling (register.php, login.php).
- **Role Management:** Differentiates administrative users (is_admin = 1) from standard students (is_admin = 0).
- **Landing Page:** Interactive homepage (index.php) featuring direct CTA buttons and automatic dashboard redirection for active sessions.
- **Account Deletion Request:** Allows users to submit structured deletion requests with justification to administrators.

5.2 Skill Management & Categorization

- **Skill CRUD Operations:** Students can create, edit, view, and delete skills they Offer to teach or Want to learn.
- **Categories:** Organizes skills under structured headings (e.g., Programming, Language, Design, Academics).
- **Tier Assignment:** Displays user skill levels (Beginner, Intermediate, Expert) based on test evaluations.

5.3 Interactive Skill Verification Engine

- **Dynamic Quiz Delivery:** Loads skill-specific questions using case-insensitive string matching and fallback question handlers.

- **Multi-Format Questions:** Supports both single-choice (radio buttons) and multiple-choice (checkboxes) options.
- **Automated Scoring & Tiering:** Score-based proficiency tier assignment (see table below).

Score Range (out of 30)	Assigned Tier
25 – 30	Expert
15 – 24	Intermediate
Below 15	Beginner

- **Score Audit Log:** Saves test attempt records (skill_test_results) and dynamically updates skill tiers on the profile.

5.4 Peer-to-Peer Skill Swapping

- **Request Pipeline:** Allows students to select an offered skill and request a wanted skill from a peer.
- **Status Lifecycle:** Tracks swap requests through Pending, Accepted, and Rejected states.

5.5 Admin Monitoring Portal

- **Analytics Cards:** Displays live counts of total registered students, shared skills, and successful swaps.
- **Verification Test Logs:** Displays student assessment history, showing score out of 30, assigned tier, skill tested, and timestamp.
- **Account Moderation:** Enables administrators to review and approve deletion requests or manage user profiles.

6. Technology Stack & System Architecture

Layer	Technology	Owner
Frontend (structure)	HTML5	Anika Tasnim
Frontend (styling)	CSS3	Abdullah As Sayef Tanzim
Backend	PHP (procedural & OOP, prepared mysqli statements)	MD. Nahidul Islam
Database	MySQL & SQL Schema Queries, XAMPP / phpMyAdmin	Nahida Islam Shormi
Environment	Apache Web Server	—

7. Installation & Setup Guide

- **Clone or Extract Project Files** — Place the project folder in your web server root (e.g., C:/xampp/htdocs/skillswap).
- **Database Import** — Open phpMyAdmin (<http://localhost/phpmyadmin>), create a database named skillswap, and import the provided schema.sql file to create tables and seed quiz questions.
- **Database Connection Setup** — Verify configuration in config/database.php:

```
<?php
$conn = new mysqli("localhost", "root", "", "skillswap");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

- **Access Platform** — Navigate to <http://localhost/skillswap/> in any web browser.

8. Conclusion & Future Enhancements

8.1 Conclusion

SkillSwap successfully provides a structured environment for peer learning. By integrating dynamic verification tests, the system removes ambiguity regarding skill proficiency levels, creating a trustworthy trading atmosphere for students.

8.2 Future Enhancements

- **Rating & Review System:** Enable students to leave star ratings and comments following completed skill swaps.
- **Real-time Chat:** Implement direct in-app messaging between swap partners.
- **Notification Engine:** Add email or push notifications for pending swap requests and test updates.